

THE GENERALIZED t -DISTRIBUTION, A GENERALIZATION TO THE POSITIVE TAIL OF THE t DISTRIBUTION

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Summary: The purpose of this paper is to introduce a distribution, which is a generalization of the positive tail of the standardized t distribution, called the Generalized t distribution. The t distribution is a well-known Pareto-type distribution which is considered in situations where the data indicates heavy or extreme tails. In some cases, as we show in this paper, it is questioned whether the tail of the t distribution is not too heavy or too light for a specific dataset. We suggest using the Generalized t distribution as an alternative distribution to model the right hand side tail in cases where the tail of the t distribution is too heavy or too light.

1. Introduction

The t distribution belongs to the class of Pareto-type distributions in Extreme Value Theory and is commonly used to model data with heavy or extreme tails, see for example Beirlant et al. (2004) and Nadarajah (2007). In some cases the t distribution seems to be a reasonable fit to the observations but closer inspection on the tail quantiles shows that the tail of the distribution is actually too heavy or too light for the specific situation. In Section 2 we consider a case where the t distribution is fitted to a real dataset. Initially the t distribution seems to fit the observations reasonably well but, after predicting future quantiles, it is evident that the tail of the t distribution is too light for the specific dataset.

In Section 3 we derive and define a distribution, which is a generalization of the positive tail of the t distribution, called the Generalized t (Gt) distribution. Section 3 shows that the Gt distribution is derived in terms of the ratio of two Generalized Gamma variables. The Gt distribution has two parameters: the shape parameter and the tail parameter (known as the tail index). The tail index is

an indication of whether the right hand side tail of the Gt distribution is heavier or lighter than the usual t distribution. Thus, the tail index gives an indication of whether the Gt distribution is more appropriate than the t distribution to model the tail observations since the tail of the t distribution might be too heavy or too light. The two parameters of the Gt distribution are estimated through a Gibbs sampling, Bayesian approach. The effectiveness of the parameter estimation method is investigated by means of a simulation study and the results are shown by means of a table. The Gt distribution is fitted to the same dataset as the t distribution in Section 2. High quantiles are predicted and compared to the quantiles predicted with the t distribution. The two distributions are compared in terms of goodness of fit through quantile-quantile (QQ) plots.

It is important to note that the purpose of this paper is to introduce and investigate the Gt distribution which is a generalization of the positive tail of the t distribution and not to compare it with other heavy tailed distributions.

2. Fitting the t distribution to a real dataset

The real dataset considered here is the total February rainfall in Bloemfontein, a city situated in the Free State province of South Africa, from 1970 to 2011. The observations are shown in Figure 1. From Figure 1 it is evident that the dataset contains large observations that indicate a heavy tail to the right. Since the t distribution belongs to the maximum domain of attraction of the extreme value distribution, with a positive extreme value index, we consider the t distribution here to model the rainfall data.

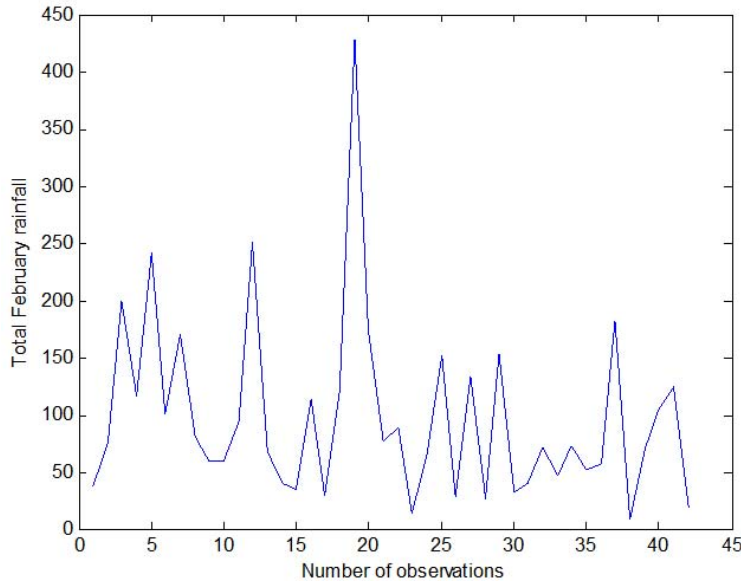


Figure 1: Bloemfontein total February rainfall for the period 1970 to 2011.

The t distribution has the following density function

$$f(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sigma} \left(\frac{1}{\pi\nu}\right)^{\frac{1}{2}} \left(1 + \frac{(x-\mu)^2}{\sigma^2\nu}\right)^{-\frac{\nu+1}{2}}, \quad -\infty \leq x \leq \infty \quad (1)$$

where μ is the location parameter, σ is the scale parameter and ν is the degrees of freedom. The t distribution is fitted to the total Bloemfontein February rainfall data (X) using the function `dffitool` in the statistical package MATLAB. The fit is shown in Figure 2.

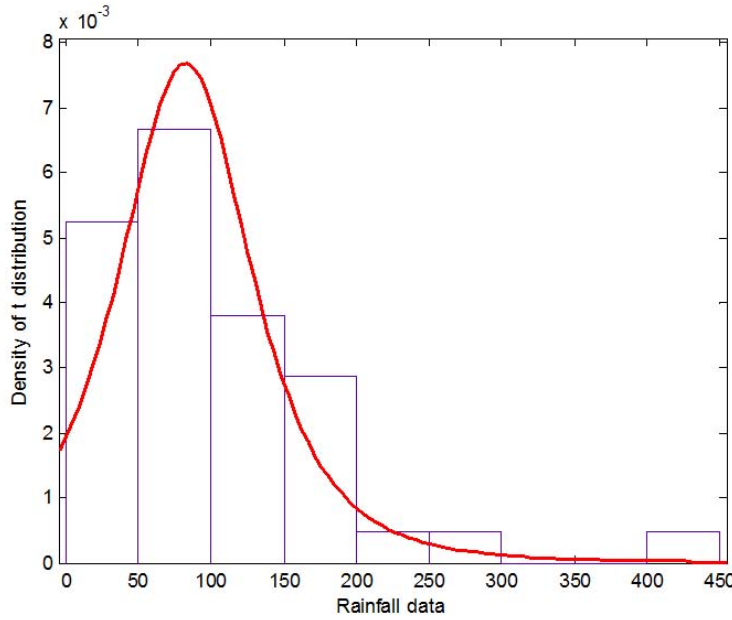


Figure 2: The t distribution fitted to the total February rainfall in Bloemfontein for the period 1970 to 2011.

The Maximum Likelihood estimates of the parameters, obtained through MATLAB, are as follows: $\hat{\mu} = 82.5567$ (10.1617), $\hat{\sigma} = 47.777$ (9.58895) and $\hat{\nu} = 2.96339$ (1.50758) with the standard errors given in brackets. From Figure 2 it seems as if the t distribution gives a reasonably good fit to the rainfall dataset. A QQ-plot (where the empirical quantiles are plotted against the theoretical quantiles of the t distribution) using the estimated parameters is shown in Figure 3. The straight line (45° line) indicates the ideal fit. If the quantiles lie close to the 45° line it indicates a good fit. From Figure 3 it seems as if the tail of the t distribution is too light to model the larger observations, especially the largest observation. The posterior predictive density is considered next for estimating high future quantiles. The posterior predictive density of the t distribution for a future observation, X_{n+1} , conditional on the data $x_1, \dots, x_n$, is as follows:

$$U^2 = \frac{(n-1)n}{(n^2-1)} \frac{(X_{n+1} - \bar{x})^2}{s^2} \sim F(1, n-1) \quad (2)$$

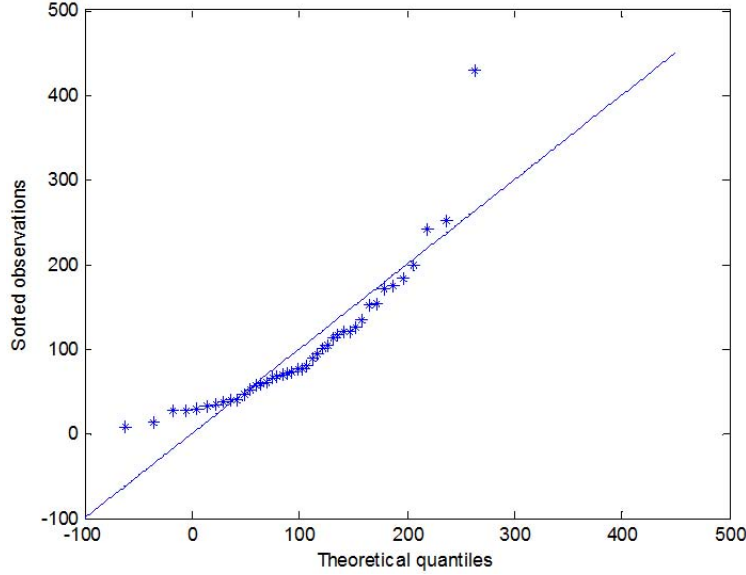


Figure 3: Empirical quantiles plotted against the theoretical quantiles.

where $s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ for $n = 42$ observations (Geisser, 1993). From Equation 2 it follows that U is standard t distributed, $U \sim t_{n-1}$. The 99th-quantile is predicted by first calculating U using the MATLAB function `tinu(0.99, n-1)`. After U is calculated it is substituted into Equation 2 and a possible future 99th-quantile, $Q_{x_{n+1}}(0.99)$, is predicted as 252.8841. The observed 99th quantile from the dataset is 428.7300, which confirms our suspicion that the tail of the t distribution is too light. The posterior predictive density for all future observations, X_{n+1} , is shown in Figure 4. Although the t distribution seems to be an appropriate distribution to model the rainfall data, as shown in Figure 2, the QQ plot and the estimated 99th-quantile indicate that the tail of the t distribution is actually too light to model the tail of the observations effectively.

This brings us to the second part of the paper. In some cases, as shown in the illustration above, the body of the t distribution seems to fit the observations well but it is questioned whether the tail of the t distribution is not too heavy or too light for a specific situation. With this in mind we derive and define a distribution, which is a generalization of the tail of the positive side of the standardized t distribution, called the Generalized t (Gt) distribution. The Gt distribution, as shown in Section 3, is defined in such a way that it allows for situations where the right hand side tail is heavier or lighter than that of the usual t distribution.

3. The Generalized t distribution

We start by deriving the Gt distribution in terms of the Generalized Gamma (GGAM) distribution. The well-known three parameter GGAM distribution is a generalization of the two parameter

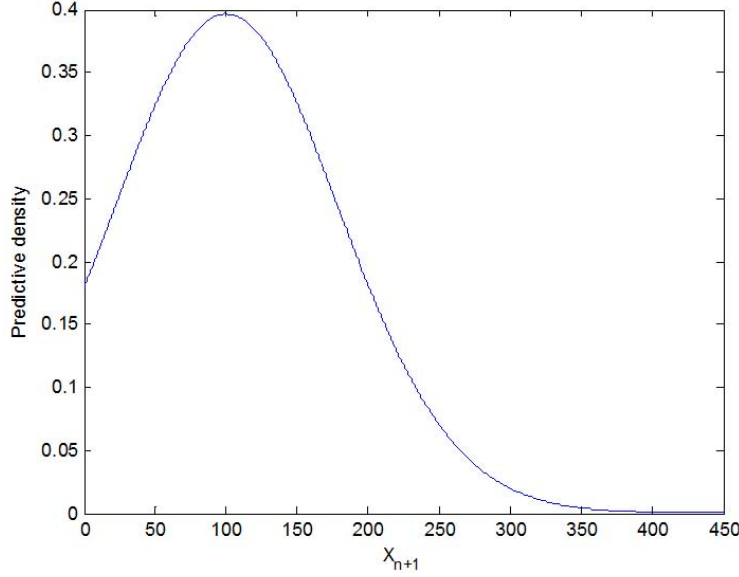


Figure 4: The posterior predictive density of the t distribution for future observations X_{n+1} .

Gamma distribution; see for example Stacy (1962).

Assume that Y is GGAM distributed. The density function of $Y \sim GGAM(k, a, \beta)$, where $k > 0$ is the shape parameter, $\beta > 0$ the tail index and $a > 0$ the scale parameter, is defined as

$$f(y | k, a, \beta) = \frac{a^k}{\beta \Gamma(k)} e^{-ay^{1/\beta}} y^{\frac{k}{\beta}-1}, y > 0. \quad (3)$$

Let $a = 1$ and assume that $Y_1 \sim GGAM(k_1, 1, \beta)$ and $Y_2 \sim GGAM(k_2, 1, \beta)$ independent of each other. It can be shown that the ratio of two Generalized Gammas, $W = \frac{Y_1}{Y_2}$, is Generalized Beta Type II (GBET2) or Generalized Beta Prime distributed, denoted by $W \sim GBET2(k_1, k_2, \beta)$, with probability density

$$f(w) = \frac{\Gamma(k_1 + k_2)}{\beta \Gamma(k_1) \Gamma(k_2)} \left(w^{\frac{1}{\beta}}\right)^{k_1 - \beta} \left(1 + w^{\frac{1}{\beta}}\right)^{-(k_1 + k_2)}, w > 0. \quad (4)$$

Assume $k_1 = \frac{1}{2}$ and make the transformation $T = \sqrt{\frac{Y_1}{Y_2}} = \sqrt{W}$ for positive $t \geq 0$. The density of T follows from (4) as:

$$f(t) = \frac{2\Gamma(\frac{1}{2} + k_2)}{\beta \sqrt{\pi} \Gamma(k_2)} \left(t^{\frac{2}{\beta}}\right)^{\frac{1-\beta}{2}} \left(1 + t^{\frac{2}{\beta}}\right)^{-(\frac{1}{2} + k_2)}, t \geq 0 \quad (5)$$

Then $T \sim Gt(k_2, \beta)$ is a Gt distribution, since (5) is a generalization of the t distribution under the transformation $t = \frac{y - \mu}{\sigma}$ in the positive domain. In the transformation y denotes the observations, μ the location parameter and σ the scale parameter of the observations where $\beta = 1$.

The mean and variance of the Gt distribution are as follows:

$$E(T) = \frac{\Gamma\left(\frac{\beta+1}{2}\right) \Gamma\left(k_2 - \frac{\beta}{2}\right)}{\Gamma\left(\frac{1}{2}\right) \Gamma(k_2)}, \quad k_2 > \frac{\beta}{2} \quad (6)$$

$$Var(T) = \frac{\Gamma\left(\frac{2\beta+1}{2}\right) \Gamma(k_2 - \beta)}{\Gamma\left(\frac{1}{2}\right) \Gamma(k_2)} - \left[\frac{\Gamma\left(\frac{\beta+1}{2}\right) \Gamma\left(k_2 - \frac{\beta}{2}\right)}{\Gamma\left(\frac{1}{2}\right) \Gamma(k_2)} \right]^2, \quad k_2 > \beta \quad (7)$$

The Extreme Value Index (EVI) of the Gt distribution follows as $\frac{\beta}{2k_2}$ (see Beirlant et al., 2004, for the EVI of $|t|$ as $\frac{1}{2k_2}$). Figure 5 shows the Gt density with $k_2 = 3$ and different values of β from 0.2 to 2 with increments of 0.2. The positive side of the standard t density (where $\beta = 1$) can be

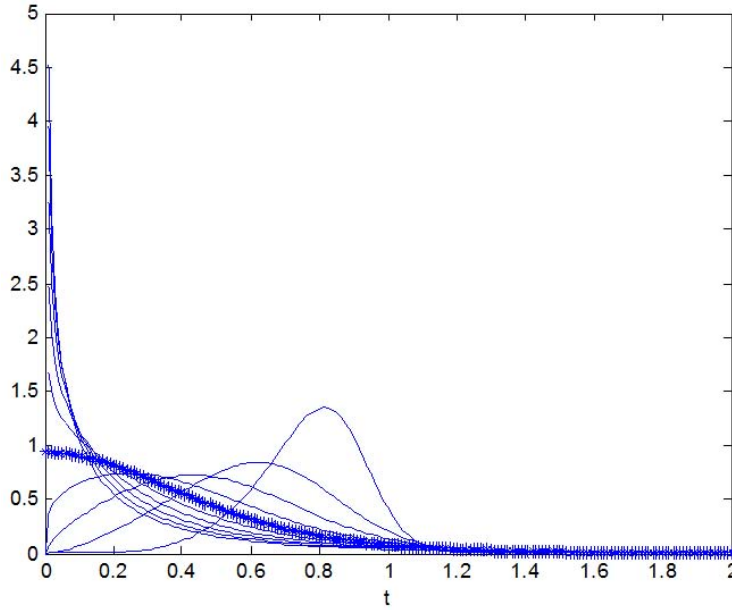


Figure 5: The Gt densities for $k_2 = 3$ and $\beta = 0.2 : 0.2 : 2$.

seen in Figure 5 printed in a thick line with (*). Specifically, it is the positive part of the symmetric density of the ordinary t_{2k_2} with $2k_2$ degrees of freedom and EVI of $\frac{1}{2k_2}$. The tail of the density becomes heavier as β increases. We can now make a very important classification and divide the Gt distribution into two groups:

- the Gt distribution with tails heavier than that of the t distribution (HTT) when $\beta > 1$ and
- the Gt distribution with tails lighter than that of the t distribution (LTT) if $\beta < 1$.

The value of the β parameter plays an important role in defining the heaviness of the tail and whether the t distribution is an appropriate distribution to model the tail. The next step is to estimate the parameters of the Gt.

3.1. Parameter estimation of the Gt distribution

A Bayesian approach is considered for estimating the two parameters of the Gt distribution jointly. The joint likelihood of k_2 and β is given as

$$like(k_2, \beta | \underline{t}) = \left(\frac{\Gamma(\frac{1}{2} + k_2)}{\beta \sqrt{\pi} \Gamma(k_2)} \right)^n \prod_{i=1}^n \left(t_i^{\frac{2}{\beta}} \right)^{\frac{1-\beta}{2}} \left(1 + t_i^{\frac{2}{\beta}} \right)^{-\left(\frac{1}{2} + k_2\right)}. \quad (8)$$

The log of the joint Maximal Data Information (MDI) prior, $\log \pi(k_2, \beta) \propto E(\log(f(T | k_2, \beta)))$, on k_2 and β under the Gt distribution is given as

$$\begin{aligned} \log \pi(k_2, \beta) &\propto \\ &\log(\Gamma(\frac{1}{2} + k_2)) - \log(\Gamma(k_2)) - \log(\beta) \\ &+ \left(\frac{1-\beta}{2}\right) E\left(\log\left(T^{\frac{2}{\beta}}\right)\right) - \left(\frac{1}{2} + k_2\right) E\left(\log\left(1 + T^{\frac{2}{\beta}}\right)\right), \quad k_2 > 0, \beta > 0 \end{aligned} \quad (9)$$

See Zellner (1977) for more information on the MDI prior. Assume that $S = T^{\frac{2}{\beta}} = \left[\left(\frac{Y_1}{Y_2} \right)^{\frac{1}{2}} \right]^{\frac{2}{\beta}} = \left(\frac{Y_1}{Y_2} \right)^{\frac{1}{\beta}}$. The last transformation indicates that S is the quotient of two Gamma random variables with parameters $(k_1 = \frac{1}{2}, 1)$ and $(k_2, 1)$ respectively. It can be shown that if X follows a Gamma distribution with parameters $(k, 1)$, $X \sim GAM(k, 1)$, then $E(\log(X)) = \psi(k)$. It now follows that $E(\log(S)) = \psi(\frac{1}{2}) - \psi(k_2)$ and $E(\log(1 + S)) = \psi(\frac{1}{2} + k_2) - \psi(k_2)$. The prior in Equation 9 can now be written as:

$$\begin{aligned} \pi(k_2, \beta) &\propto \frac{\Gamma(\frac{1}{2} + k_2)}{\beta \Gamma(k_2)} \exp \left\{ \left(\frac{1-\beta}{2} \right) \left(\psi\left(\frac{1}{2}\right) - \psi(k_2) \right) - \left(\frac{1}{2} + k_2 \right) \left(\psi\left(\frac{1}{2} + k_2\right) - \psi(k_2) \right) \right\}. \end{aligned} \quad (10)$$

The joint posterior distribution of k_2 and β is then constructed as

$$\pi(k_2, \beta | \underline{t}) \propto like(k_2, \beta | \underline{t}) \pi(k_2, \beta). \quad (11)$$

To simulate a set of (k_2, β) 's from the posterior, Equation 11, we use the Gibbs sampling method. This involves simulating alternatively k_2 from its conditional density function given a fixed β and then β from its conditional density given the selected k_2 . This process is repeated with a large number of iterations.

3.1.1. Simulation study

In this section we investigate the effectiveness of the parameter estimation method, discussed in Section 3.1 above, by means of a simulation study. A number of Gt values are generated for various combinations of β and k_2 values. For each of the generated Gt datasets a set of (k_2, β) pairs is simulated from the posterior distribution using the Gibbs sampling method discussed above, with 1000 iterations. The estimated parameter values are obtained by calculating the means of the 1000 simulated k_2 's and β 's respectively. The number of generated Gt values (sample size) varies between 100 and 1000 with increments of 100. The results of the following combinations of β and k_2 values:

$(\beta = 0.2, k_2 = 2)$; $(\beta = 0.5, k_2 = 2)$; $(\beta = 0.9, k_2 = 2)$ are shown in Table 1 in the Appendix. Various other combinations were also considered, although it is not shown here. Column 1 shows the true parameter values and the various samples sizes, Columns two and four show the Bayesian estimated parameter values, the means of the Gibbs realizations. From Table 1 one can see that the estimated parameter values are close to the true parameter values for all the different combinations of β and k_2 . Columns three and five gives the mean squared error difference between the 1000 Bayesian simulated parameter values and the true parameter values. The small mean squared error differences indicate little variability showing that our method is appropriate for estimating the parameters.

We continue to check the effectiveness of our estimation method by comparing the Maximum Likelihood estimate (MLE) of the degrees of freedom parameter, ν , of the standardized t distribution with the Bayesian parameter of the degrees of freedom, $2k_2$. Let $T \sim Gt(k_2, \beta)$ then $Z = T^{\frac{1}{\beta}}$ follows the distribution of the positive part of a standardized t distribution with $2k_2$ degrees of freedom, $Z \sim |t_{2k_2}|$. After estimating the two parameters of the Gt distribution the above transformation to the t distribution is made and the degrees of freedom parameter, ν , is estimated through maximum likelihood by using the statistical package MATLAB. The MLE is obtained for different sample sizes and different combinations of β and k_2 and shown in Column 7 of Table 1. The MLE of ν is compared with Column 6 of Table 1 which is the Bayesian estimate of the degrees of freedom, $2\hat{k}_2$. The values in Columns 6 and 7 are close. Column 8 shows the mean squared error difference between the 1000 Bayesian simulated $2k_2$ parameter values and the true parameter value of ν . The relatively small values in Column 8 indicate that our method of estimation is appropriate.

3.2. High quantile predictions

The posterior predictive density of a future observation T_{n+1} , given the data, is given as follows (see for example Gamerman and Lopes, 2006):

$$p(t_{n+1}|\underline{t}) \propto E_{k_2, \beta|\underline{t}} f(t_{n+1}|k_2, \beta). \quad (12)$$

Equation 12 cannot be solved explicitly, but the posterior predictive density is simulated by taking the mean of the densities at t_{n+1} for a large number of k_2 and β values simulated from the posterior. Equation 12 is estimated as

$$\hat{p}(t_{n+1}|\underline{t}) = \frac{1}{m} \sum_{j=1}^m f(t_{n+1}|k_{2j}, \beta_j) \quad (13)$$

where m is the number of simulated k_2 and β pairs.

The posterior predictive quantile function, Equation 14, can now be used to predict a high p^{th} quantile

$$Q_p(\underline{t}) \propto E_{k_2, \beta|\underline{t}} Q_p(t_{n+1}|k_2, \beta). \quad (14)$$

Equation 14 is again estimated by doing a similar simulation where a number of k_2 and β pairs are simulated from the posterior and substituted into Equation 15

$$\hat{Q}_p = \frac{1}{m} \sum_{j=1}^m Q_p(t_{n+1}|k_{2j}, \beta_j). \quad (15)$$

The 99th-quantile, for example, can easily be estimated through the statistical package MATLAB by generating a 99th quantile from a Beta distribution with parameters $(\frac{1}{2}, k_2)$ using the function $q_{Beta} = \text{betainv}(0.99, \frac{1}{2}, k_2) \cdot q_{Beta}$ is then transformed to a Beta Type II quantile as follows: $q_{BetaT2} = \frac{q_{Beta}}{1 - q_{Beta}}$. Finally a Gt quantile is obtained through the following transformation: $q_{Gt} = q_{BetaT2}^{\frac{\beta}{2}}$. The mean of the q_{Gt} 's is an estimate of the predicted quantile. Since T is a standardized variable, the quantile must be rescaled, $q_{Gt}\sigma + \mu$, to obtain the true estimated quantile.

3.3. Fitting the Gt distribution to a real dataset

The Gt distribution is fitted to the positive standardized total Bloemfontein February rainfall data, the same dataset we considered in Section 2. The two parameters of the Gt distribution are estimated through the Gibbs sampling method as discussed in Section 3.1. Figure 6 shows a scatter plot of the simulated k_2 's against β 's for the total Bloemfontein February rainfall data. The means of the simulated k_2 's and β 's are calculated as 1.2846 and 1.2582 respectively and are considered as the estimates of k_2 and β respectively. Note that $\beta > 1$ indicates a heavier tail than the t distribution. We therefore suggests that the t distribution is not appropriate for estimating high total February rainfall tail quantiles, since the tail of the t distribution is too light. The Gt distribution will be more suitable.

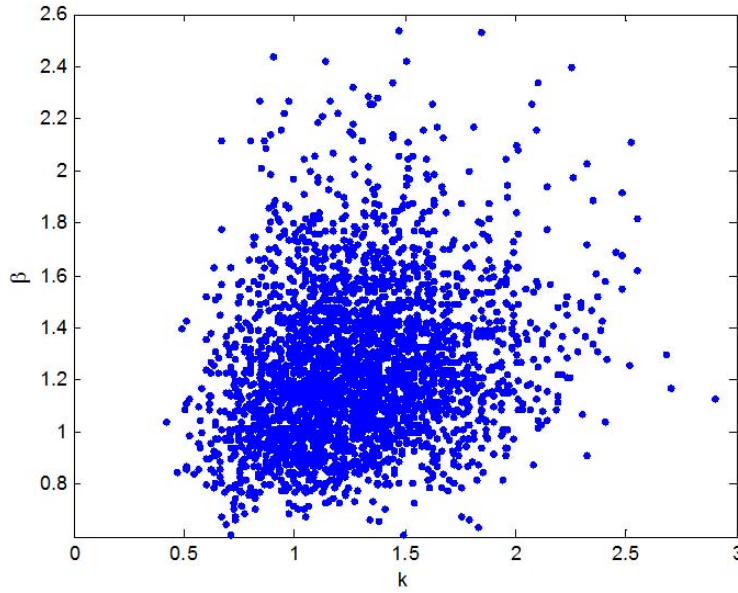


Figure 6: Plot of 2000 simulated (k_2, β) values from the Gibbs sampler.

A QQ-plot of the Gt distribution with the estimated parameters are constructed and shown in Figure 7 where the straight line indicates the 45° line. Figure 7 gives us an indication of the goodness of fit of the Gt distribution to the observations. From Figure 7 it seems that although the tail of the Gt distribution is still too light to model the larger observations it is closer than the t distribution.

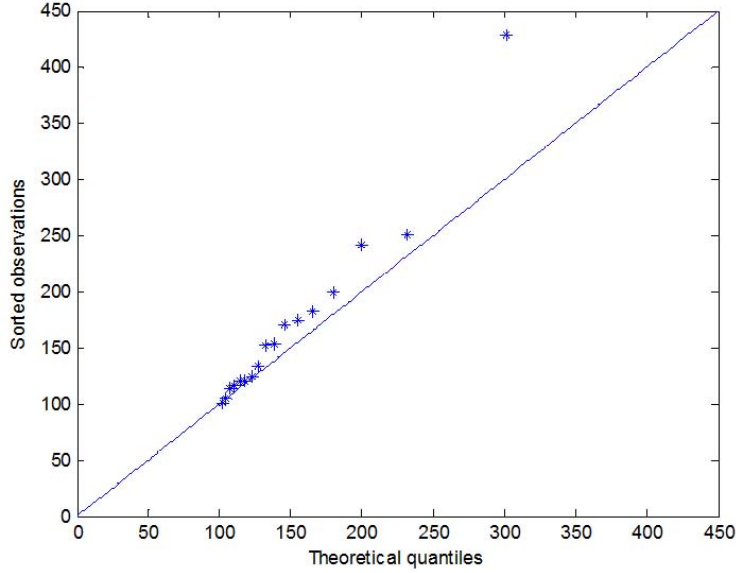


Figure 7: Empirical quantiles plotted against the theoretical quantiles.

The posterior predictive density of the Gt distribution for future observations is shown in Figure 8. The 99th-quantile is predicted as explained in Section 3.2 as $q_{Gt} = 3.3230$ which is then rescaled to: $q_{Gt}\sigma + \mu = 312.6871$. The observed 99th quantile is 428.7300. Although the predicted 99th-quantile of the Gt distribution is smaller than the observed quantile it is still much larger than the quantile predicted with the t distribution. This was expected since $\beta > 1$ indicates that the tail of the Gt distribution is heavier than the t distribution and thus in this illustration the Gt distribution will be a more appropriate distribution to model the data.

4. Conclusion

In this paper we introduced the Gt distribution, a generalization of the positive side of the tail of the standardized t distribution. We showed that the Gt distribution can be used to model data with a heavier or lighter right hand side tail than what you would expect from the t distribution. The following observation was made about the β parameter (tail index) of the Gt distribution: $\beta < 1$ indicates a lighter tail than the t distribution and $\beta > 1$ indicates a heavier tail than the t distribution. The two parameters of the Gt can be estimated through Gibbs sampling when considering a Bayesian approach. The posterior predictive density and posterior predictive quantiles can also be predicted, as shown in Section 3. The Gt and t distributions were compared through means of an illustration on the total February Bloemfontein rainfall data. The β parameter was estimated to be larger than one, thus indicating that the right tail of the Gt distribution is heavier than that of the t distribution. This was confirmed when the high quantiles were predicted. The predicted 99th quantile from the Gt distribution was closer to the observed quantile than the predicted 99th quantile of the t distribution.

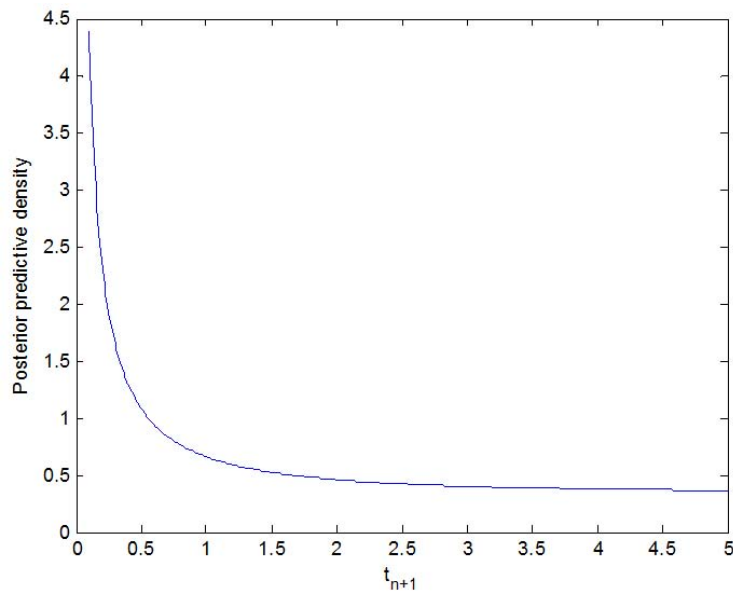


Figure 8: The posterior predictive density of the Gt distribution for future observations T_{n+1} .

Although the Gt distribution is not a perfect fit to the dataset it definitely performs better than the t distribution since it allows for heavier tails.

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Appendix

Table 1: Simulation study on the Bayesian parameter estimation method.

$\beta = 0.2, k_2 = 2$	$\hat{\beta}$	$MSE(\hat{\beta})$	$\hat{k}_2$	$MSE(\hat{k}_2)$	$2\hat{k}_2$	ν	$MSE(2\hat{k}_2)$
$n = 100$	0.2047	0.0003	2.1259	0.0757	4.2518	3.6627	0.6106
200	0.2098	0.0003	1.7605	0.0347	3.5209	3.4338	0.0975
300	0.2215	0.0006	1.8484	0.1195	3.6969	2.8570	0.7774
400	0.2014	0.0001	2.0374	0.0247	4.0749	4.5492	0.2959
500	0.2144	0.0003	2.0375	0.0139	4.0749	3.2660	0.7098
600	0.2	0.00004	1.9209	0.0281	3.8418	3.2474	0.3918
700	0.1913	0.0001	1.9837	0.0090	3.9675	3.7457	0.0849
800	0.2138	0.0002	2.0638	0.0305	4.1276	3.2498	0.8069
900	0.1962	0.00004	2.0517	0.0084	4.1035	4.4768	0.1698
1000	0.1992	0.00003	2.1357	0.0402	4.2713	4.9388	0.4762

$\beta = 0.5, k_2 = 2$	$\hat{\beta}$	$MSE(\hat{\beta})$	$\hat{k}_2$	$MSE(\hat{k}_2)$	$2\hat{k}_2$	ν	$MSE(2\hat{k}_2)$
$n = 100$	0.4741	0.0021	2.3630	0.0973	4.7259	4.6217	0.3878
200	0.4749	0.0014	1.9729	0.0646	3.9458	4.6749	0.6728
300	0.5348	0.0018	1.6949	0.0524	3.3898	2.7838	0.4294
400	0.5410	0.0021	2.2125	0.0685	4.4250	3.7827	0.4990
500	0.5056	0.0004	2.0536	0.0164	4.1071	3.9366	0.0897
600	0.5179	0.0006	2.0283	0.0120	4.0565	3.4677	0.3887
700	0.5301	0.0012	1.9255	0.0133	3.8510	3.5114	0.1494
800	0.5065	0.0002	2.0577	0.0090	4.1154	4.1194	0.0359
900	0.5096	0.0003	2.2231	0.0114	4.4463	4.0053	0.2328
1000	0.5037	0.0002	2.0709	0.0098	4.1418	4.0768	0.0314

$\beta = 0.9, k_2 = 2$	$\hat{\beta}$	$MSE(\hat{\beta})$	$\hat{k}_2$	$MSE(\hat{k}_2)$	$2\hat{k}_2$	ν	$MSE(2\hat{k}_2)$
$n = 100$	0.8564	0.0067	2.3012	0.1626	4.6024	4.5511	0.3457
200	0.9940	0.0121	1.9478	0.0731	3.8957	3.6840	0.1705
300	0.9036	0.0016	2.0626	0.0241	4.1252	3.6603	0.3035
400	0.8863	0.0014	2.0410	0.0179	4.0821	4.4944	0.2402
500	0.9424	0.0029	2.0152	0.0438	4.0303	3.7259	0.1478
600	0.9102	0.0010	2.1423	0.0167	4.2846	4.9634	0.5107
700	0.8995	0.0007	2.2208	0.0152	4.4417	4.2640	0.0784
800	0.8869	0.0008	1.9722	0.0191	3.9444	3.9563	0.0319
900	0.8951	0.0006	2.0869	0.0105	4.1738	4.6505	0.2586
1000	0.9504	0.0032	1.8557	0.0147	3.7115	3.3433	0.1565